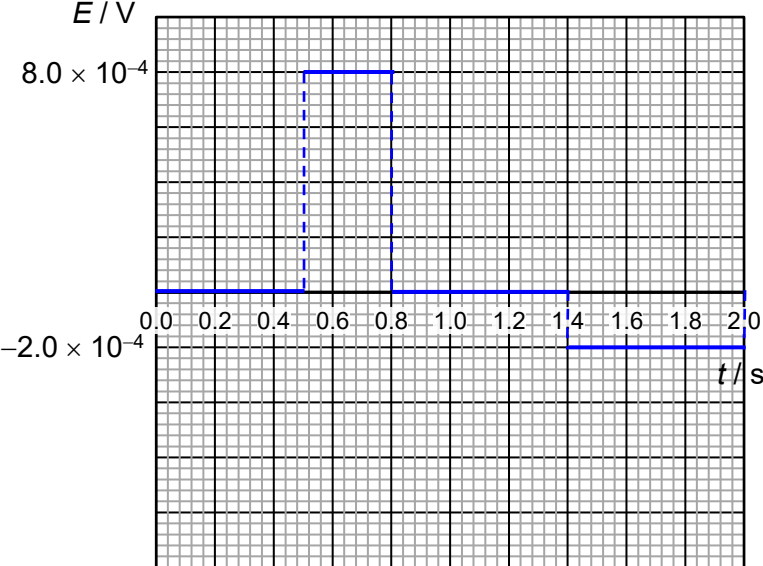


		velocity
6(a)	Faraday's law of electromagnetic induction states that the induced e.m.f. in a coil is directly proportional to the rate of change of magnetic flux linkage through the coil.	[1]
6(b)(i)	$B = \mu_0 n I$ $= 4\pi \times 10^{-7} \times \left(\frac{400}{1.6} \right) \times 3.8$ $= 1.19 \times 10^{-3} \text{ T}$ $\Phi = NBA$ $= 80 \times (1.19 \times 10^{-3}) \times \pi \times \left(\frac{0.040}{2} \right)^2$ $= 1.2 \times 10^{-4} \text{ Wb}$	 [1] value of B [1] sub [1] ans
6(b)(ii)	$E_{\text{mean}} = \frac{\Delta \Phi}{\Delta t}$ $= \frac{2 \times 1.2 \times 10^{-4}}{0.30}$ $= 8.0 \times 10^{-4} \text{ V}$	 [1] sub [1] ans

No.	Solution	Remarks
6(b)(iii)		[1] $E = 0$ V from $t = 0$ to 0.5 s & 0.8 s to 1.4 s [1] $E = 8.0 \times 10^{-4}$ V from $t = 0.5$ s to 0.8 s [1] $E = -2.0 \times 10^{-4}$ V from $t = 1.4$ s to 2.0 s
6(b)(iv)	The magnitude of the <u>maximum e.m.f. induced in the coil increases</u>, The iron core in the solenoid <u>increases the magnetic flux density in the coil</u>, resulting in <u>increase in the magnetic flux linkage in the coil</u>. Hence by Faraday's law, the magnitude of the maximum e.m.f. induced in the coil increases,	[1] state [1] explain